Importance of Quality Criteria for Video Tutorials

Which Factors Influence the Learner Experience in Video Tutorials?

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***Abstract*—The paper investigates the relative importance of quality criteria for video tutorials. Video tutorials are a widely used method that enables learners to acquire new skills at their own pace, making them popular across various domains, including commercial applications. However, the quality of a tutorial depends on its design and its alignment with learners' needs. In this exploratory study, we reveal considerable differences in how these criteria are valued by learners and identify the aspects most critical for the learner experience. Furthermore, we demonstrate that the type of tutorial influences the perceived importance of these criteria. These insights offer practical guidance on enhancing the quality of video tutorials. In addition, they help to define which questions should be asked in questionnaires designed to assess learner experience with these tutorials.**

Keywords-*Video tutorials; Tutorial quality; Importance of quality aspects; Learner experience.*

I. INTRODUCTION

Tutorials break down complex topics into clear, manageable steps. They often include practical examples and exercises that help to test the acquired knowledge. This makes them a popular method for learning. Tutorials are available in many formats. In particular, online tutorials or simple video tutorials are easily accessible.

There are many different applications for online tutorials [1]. For example, tutorials that explain how to operate (e.g., use a TV remote control), maintain (e.g., change the SIM card of a smartphone), or repair (e.g., replace defective parts) technical devices. Tutorials are also frequently used to explain how to do special tasks in common software products (e.g., perform not fully intuitive tasks with MS Excel or MS Word) and in some sense act as a potential alternative to classical online documentation [4]. They are also quite popular to teach how to use programming environments (e.g., tutorials concerning web development like SelfHTML) [5].

In industry, tutorials are used by companies to transfer knowledge to customers to boost sales and reduce service costs [7], by educational or scientific organizations to teach important concepts [2], or by individual contributors who simply want to share their knowledge.

The term tutorial is often used with various meanings. Within this paper, we define tutorials as instructional materials that teach specific skills or the ability to perform specific tasks in a sequence of steps without the assistance of an instructor or the ability to ask questions in a community.

In this article, we address two research questions: RQ1: Which quality criteria are most important for the learner experience with video tutorials? RQ2: Does the perceived importance of these criteria depend on the type of tutorial? We demonstrate that different types of video tutorials, categorized into four classes, need to be evaluated against different quality criteria. Our findings show that no universal set of quality criteria applies equally to all tutorial types.

The paper is structured as follows. In Section II, we discuss tutorial quality, and in Section III, related work. Section IV introduces our classification of video tutorial types, Section V describes the study design, and Section VI presents the results. Section VII discusses limitations, and Section VIII concludes the paper.

II. TUTORIAL QUALITY

A tutorial needs to fulfill certain quality criteria to effectively support learners. Tutorials must be well-designed, engaging, and easy to follow. This is especially important in commercial contexts. Badly designed tutorials can have a negative impact on the market success or reputation of a company. We describe this in a few examples.

Customers often use tutorials for software in combination with free trial versions to decide if they want to purchase a product or service or not. In such cases, a tutorial is often the first contact of a customer with a vendor and is a cheap and powerful marketing tool, but if such tutorials are not well-designed, customers may not purchase the tested product [3][12][13] and even have a bad impression of the vendor.

Another important use case of tutorials is customer support [7]. If customers have issues with a product and do not want to call support, they often search for tutorials online. If the tutorial helps to solve the issue, the customer gets quick help, and the vendor saves resources on customer support. If not, this can cause frustration and loss of loyalty.

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In principle, the criteria for tutorial quality do not differ between commercially used tutorials and tutorials provided by individuals without any commercial interest. However, the consequences when tutorial quality is not met and the resources to measure this quality are different. Due to the possible negative impact, a measurement of tutorial quality is mainly of interest in the commercial sector.

Different learners will have varying opinions regarding the quality of a tutorial. The perception of learner experience is therefore inherently subjective, similar to the concept of User Experience (UX) [9]. Measuring tutorial quality requires collecting feedback from larger target groups. This makes online questionnaires an efficient tool for this purpose. Such questionnaires can be triggered when a learner completes the tutorial or reaches a certain point within the tutorial flow [8].

III. Related Work

Quality criteria for tutorials share similarities with UX quality aspects of applications [9]. These UX quality aspects describe user needs or expectations regarding using interactive systems. Similarly, quality criteria for tutorials describe learners' needs while using a tutorial to acquire knowledge. Thus, some basic insights concerning UX quality aspects can be assumed to also apply to tutorial qualities.

An important point repeatedly found for UX qualities [10] is that their importance concerning the overall UX impression depends mainly on the type of application and the typical usage scenarios. For an overview of several studies concerning this point, see [10].

Transferring this finding to tutorials, it can be assumed that the tutorial type, defined by the main motivation of people to use it for knowledge acquisition, has an impact on the importance of quality aspects for the overall perception of or satisfaction with the tutorial. Therefore, we investigate this aspect for different types of tutorials.

This paper relates to some previous results concerning general tutorial quality aspects. In [8], a questionnaire concerning online tutorials was constructed. In that study, 117 participants rated 7 online tutorials with a set of 22 candidate items. A principal component analysis extracted two potential scales, called Structural Clarity (the tutorial is well-structured and contains only relevant content) and Transparency (the tutorial is transparent concerning the content and the learning goals). In a follow-up study [11] using a broader spectrum of tutorials, three additional scales were uncovered: Stimulation (it is interesting and motivating to work through the tutorial), Content Quality (the content, i.e., examples, illustrations, or demonstrations, is relevant and of high quality) and Understandability (the content of the tutorial is explained in easy-to-understand language). Some of the tutorial qualities used in this paper clearly refer to these scales, but we included additional qualities to check whether there is a need to add scales to the current portfolio to fully cover the concept of learner experience with tutorials.

IV. Different Types of Video Tutorials

It is obvious that various types of video tutorials exist. However, to the best of our knowledge, there is currently no established or standardized classification of tutorial types. We propose a classification framework that categorizes video tutorials along three dimensions, each representing a continuum between two extremes. These dimensions can be more clearly defined based on the tutorial's design and content characteristics:

Incidental (1–) vs. Intentional Learning (1+)
Incidental learning occurs when users directly apply the information shown in a tutorial without expecting to retain it long-term, as they do not anticipate repeating the task (e.g., "making popcorn" or "wrapping a gift"). Intentional learning occurs when users aim to master the content for repeated application (e.g., learning Excel form creation or PowerPoint master slides for professional use).

Direct Instruction (2–) vs. Exploratory (2+)
This didactic dimension depends on the learner's prior knowledge and the tutorial's learning objectives. Direct instruction provides step-by-step guidance (e.g., mixing a Mojito following precise measurements, or tying a bowline knot with specific steps), while exploratory tutorials encourage discovery and experimentation (e.g., exploring different tea recipes or photo editing techniques).

Goal-Oriented (3–) vs. Experience-Oriented (3+)
Goal-oriented tutorials focus on achieving a specific outcome efficiently (e.g., "creating an Excel form" or "learning the bowline knot"). Experience-oriented tutorials emphasize the process and broader understanding (e.g., "exploring professional cutting techniques" or "learning perspective drawing for creative expression"). This dimension is influenced by both the content and the user's motivational context.

V. A Study on the Importance of Quality Criteria for Tutorials

We investigate the importance of common quality criteria for four different types of video tutorials.

A. Experimental Design

The three dimensions of our classification framework create a 2×2×2 factorial space with eight possible combinations. Using a fractional factorial design (2^(3-1) with generator z = x·y) [14], we can efficiently cover the entire space with only four tutorial groups. This design balances all main effects while aliasing interaction effects, an acceptable trade-off for validating the classification framework. The four groups occupy opposite corners of the factorial cube, maximizing variance: Group 1 (1–, 2–, 3+), Group 2 (1–, 2+, 3–), Group 3 (1+, 2–, 3–), and Group 4 (1+, 2+, 3+).

The study used a within-subject design, i.e., each participant completed all four tutorial groups in four rounds, one tutorial per round. Within each group, participants chose one of three thematically fitting tutorials, allowing individual preference while maintaining the experimental structure.

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Participants were invited by email. The email contained a description of the overall approach of the study and links to all offered tutorials and their grouping (see Appendix A). In addition, a link to a survey was offered for the four rounds of the study. Participants had 4 weeks to finish the four rounds.

B. Presented Tutorials

Twelve video tutorials were selected for the study. They belong to the tutorial types described above. Each tutorial type was represented by 3 of the 12 tutorials. The selection of the tutorials was exploratory and followed two practical criteria: the topics should appeal to the student sample, so that participants would be likely to watch the videos completely, and the set of tutorials should cover the assumed dimensions of the classification framework as broadly as possible. The following tutorials were used across the four groups:

Group 1 (1–, 2–, 3+): Incidental Learning, Direct Instruction, Experience-Oriented

- **Mojito Cocktail:** Step-by-step instructions for mixing a classic Mojito cocktail
- **Video Editing in Canva:** Introduction to basic video editing features in Canva
- **Making Popcorn:** Guide to preparing popcorn in a pot on the stove

Group 2 (1–, 2+, 3–): Incidental Learning, Exploratory, Goal-Oriented

- **Professional Photo Editing:** Quick techniques for editing photos with various tools and effects
- **Four Best Tea Recipes:** Exploration of different tea preparation methods and flavor combinations
- **Gift Wrapping for Beginners:** Various creative approaches to wrapping gifts

Group 3 (1+, 2–, 3–): Intentional Learning, Direct Instruction, Goal-Oriented

- **Excel Form Creation:** Detailed instructions for creating a simple form in Excel
- **PowerPoint Master Slides:** Tutorial on working with master slides in PowerPoint
- **Bowline Knot:** Systematic demonstration of tying a bowline knot for sailing

Group 4 (1+, 2+, 3+): Intentional Learning, Exploratory, Experience-Oriented

- **Professional Cutting Techniques:** Exploration of knife skills and cutting methods in cooking
- **Perspective Drawing:** Learning to draw objects in perspective with various techniques
- **Creating Custom Fonts:** Creative process of designing and creating a personalized typeface

C. Participants

Participants were recruited from a course at the University of Applied Sciences Emden/Leer in the winter semester 2025. A total of 43 students participated in the study (22 male, 21 female). The average age of participants was 31.16 years (standard deviation of 6.46 years). Participants received course credits for submitting their evaluations.

D. Design of the Survey

The survey contained at the top a short instruction (translated from German):

You have watched one of the suggested tutorials. In this questionnaire, we would like to find out how important certain characteristics of a tutorial of this type are to you. It is not about whether the tutorial you watched actually has these characteristics! We are interested in how important it would be for a good learning experience if the tutorial had these characteristics. Of course, which characteristics are important depends on the type of tutorial. Therefore, please assess the importance of these characteristics regarding the tutorial you watched, and not in general.

Participants could specify in each round the tutorial they watched from a drop-down list. This was followed by questions concerning gender and age of the participants. Existing knowledge was captured with the question *How much prior knowledge do you have about the subject covered in the tutorial?* with the 5 answer categories (*Very little, Little, Some, Much, Very much*). Interest in the topic of the tutorial was addressed by the question *How interesting do you find the topic presented in the tutorial?* with the answer categories *(Very uninteresting, Uninteresting, Neither uninteresting nor interesting, Interesting, Very interesting)*.

The 17 items were derived from the item pools of the previous studies [8][11] and extended by additional candidate criteria, as described in Section III. Items concerning the importance of quality aspects (the ID before the item text is used later to refer to a specific item):

- I1: How important is it that a tutorial of this type is structured into clear, logical steps?
- I2: How important is it that a tutorial of this type clearly describes the prerequisites and expected outcomes at the beginning?
- I3: How important is it that a tutorial of this type is visually appealing?
- I4: How important is it that a tutorial of this type motivates further engagement with the topic?
- I5: How important is it that a tutorial of this type uses practical examples and appropriate visualizations to clearly explain the content?
- I6: How important is it that a tutorial of this type contains easily understandable explanations?
- I7: How important is it that a tutorial of this type has a pace that is appropriate for me?
- I8: How important is it that a tutorial of this type contains only relevant content and has a satisfactory duration?
- I9: How important is it that the information provided in a tutorial of this type is up to date?

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- I10: How important is it that a tutorial of this type not only explains how something is done, but also why it is done that way?
- I11: How important is it that a tutorial of this type also describes possible problems and offers troubleshooting tips in case something goes wrong?
- I12: How important is it that a tutorial of this type motivates you to experiment and develop your own creative ideas?
- I13: How important is it that the learner in a tutorial of this type can skip uninteresting parts or directly access interesting parts?
- I14: How important is it that the amount of information presented in a tutorial of this type is manageable at all times?
- I15: How important is it that all steps described in a tutorial of this type work exactly as described in the tutorial?
- I16: How important is it that a tutorial of this type offers concrete exercises to apply what has been shown?
- I17: How important is it that a tutorial of this type provides immediate, helpful feedback on the learner's solutions or steps?

Items could be rated on a 7-point answer scale. Thus, the presentation of an item in the survey looks like this:

How important is it that a tutorial of this type is structured into clear, logical steps?

Not important o o o o o o o *Very important*

Answers were scored from -3 to +3, starting from left to right. Of course, all quality criteria describe positive aspects of a tutorial.

VI. RESULTS

Overall, 173 ratings were received (some participants rated 5 and some 3 instead of the requested 4 tutorials).

First, we check if gender, interest in the topic, or existing knowledge concerning the topics influenced the importance ratings.

A series of Welch t-tests showed that gender had a significant impact on importance only for three of the 17 quality criteria. The impact was found for I2 (t: 3.192, p: 0.002, df: 144), I7 (t: 2.173, p: 0.031, df: 159) and I8 (t: 2.257, p: 0.025, df: 161).

Concerning prior knowledge, the numbers of participants who chose a category are (in parentheses): Very little (29), Little (42), Some (61), Much (32), Very much (9).

An Analysis of Variance (ANOVA) showed that prior knowledge impacted only the two criteria I9 (F: 2.783, p: 0.029, n: 167) and I10 (F: 3.245, p: 0.014, n: 167).

Concerning interest in the topic of a tutorial, the observed numbers are: Very uninteresting (2), Uninteresting (8), Neither uninteresting nor interesting (36), Interesting (98), Very interesting (29).

Interest in the topic had a significant impact on the importance rating of I3 (F: 3.374, p: 0.011, n: 167), I8 (F: 4.025, p: 0.004, n: 167), I13 (F: 2.9, p: 0.023, n: 167), and I15 (F: 2.757, p: 0.03, n: 167).

Overall, these attributes had only a small impact on some importance ratings. However, in real use cases, people will usually only watch tutorials if they have some interest in the topic and are not already experts, i.e., if they do not have too much prior knowledge. The observed data concerning prior knowledge and interest indicate that the tutorials offered meet these criteria, i.e., it is not unlikely that participants would watch such tutorials outside the study.

While the preceding checks used inferential tests, the following analysis is exploratory and mainly descriptive. We now look at the mean importance of the 17 quality criteria across all data, i.e., the ratings obtained for all rated tutorials independent of their group. Table I shows the mean and standard deviation of the ratings.

TABLE I. MEAN AND STANDARD DEVIATION (IN PARENTHESES) OF THE IMPORTANCE RATINGS OVER ALL RATED TUTORIALS. THE ITEM LABELS I1–I17 CORRESPOND TO THE CRITERIA SHOWN IN FIGURE 1.

ID	Mean (Std.Dev)	ID	Mean (Std.Dev)
I1	2.13 (1.20)	I10	0.43 (2.00)
I2	1.47 (1.59)	I11	0.28 (1.88)
I3	0.63 (1.66)	I12	-0.7 (2.00)
I4	-0.18 (1.95)	I13	0.32 (1.83)
I5	2.17 (1.24)	I14	1.14 (1.58)
I6	2.24 (0.86)	I15	1.98 (1.35)
I7	1.63 (1.42)	I16	-1.06 (2.00)
I8	1.59 (1.48)	I17	-1.76 (2.02)
I9	0.05 (2.34)		

If we look at the importance ratings shown in Figure 1, only 8 of the 17 quality criteria reach the "practically relevant importance level +1" over all tutorials. Due to the positive formulation of all items, a general tendency towards values above the neutral point can be expected. Therefore, values at or below +1 indicate that a criterion is not considered practically relevant for the overall perception and evaluation of a tutorial.

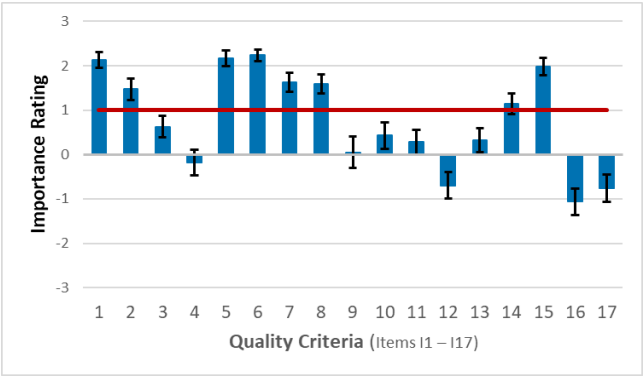


Figure 1. Mean importance ratings over all rated tutorials for quality

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criteria I1–I17. Ratings above +1 indicate practically relevant importance. N = 173 ratings.

We can also see in Table I that those criteria that exceed the threshold typically show a lower standard deviation concerning the rating, i.e., the agreement of participants concerning their importance is a bit higher in those cases.

We investigate now if the importance ratings depend on the tutorial group. Figure 2 shows the mean importance ratings per tutorial group; the exact values are listed in Table II in Appendix C.

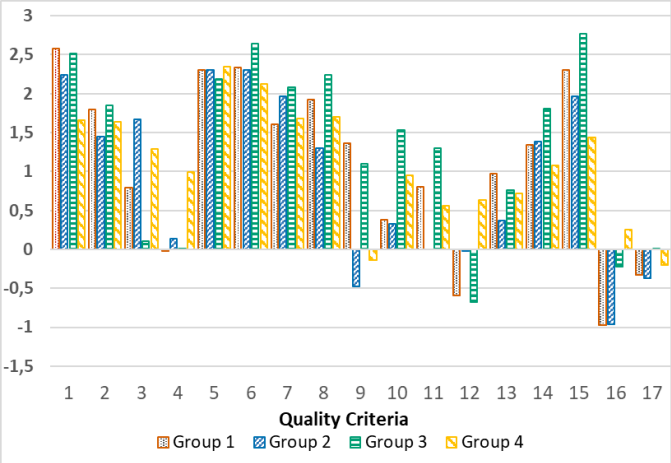


Figure 2. Mean importance ratings per tutorial group (G1–G4) for quality criteria I1–I17. N = 173 ratings.

As we can see in Figure 2, most of the quality criteria show a uniformly high or low score for the different groups. But there are some exceptions. For example, I3 and I10 reach the threshold of 1 for some tutorial groups and are clearly below it for others.

VII. LIMITATIONS

Our results are based on a sample of German students. Thus, evidently our participants are younger than the average population and have a higher level of education. In addition, cultural differences might cause different results in target groups selected from different cultures, and the selection of the 12 tutorials offered might impact the results. Moreover, the sample size (43 participants providing 173 ratings) limits the robustness of the results; the observed differences should therefore be interpreted as tendencies rather than confirmed effects.

However, as outlined above, the quality aspects for tutorials show similarities to UX aspects [9]. For UX aspects, it is known that their subjective importance is mainly influenced by the product category and the typical use cases performed with the product [6]. Cultural background or age of users play only a minor role in this area. In addition, the importance ratings remain relatively consistent across different products in a product category [10]. We assume that this also holds for quality aspects of tutorials; the results should generalize to other groups of learners and similar types of video tutorials. However, this assumption should be verified empirically. In particular, future studies should investigate whether learners from different age groups perceive the importance of these quality criteria differently.

VIII. CONCLUSION AND FUTURE WORK

We investigated which quality aspects are important for video tutorials. Based on the 173 user judgements, we identified the following 8 aspects as showing sufficiently high ratings to be considered relevant:

- The tutorial should be structured into clear, logical steps.
- The tutorial should clearly describe the prerequisites and expected outcomes at the beginning.
- The tutorial should use practical examples and appropriate visualizations to clearly explain the content.
- The tutorial should contain easily understandable explanations.
- The tutorial should present content at an appropriate pace.
- The tutorial should only contain relevant content and have a satisfactory duration.
- The amount of information presented in a tutorial should be manageable at all times.
- All steps in a tutorial should work exactly as described.

These points can guide authors in the design and testing of their tutorials. In addition, they indicate which aspects of a tutorial interaction should be measured in a corresponding questionnaire used for quality control.

Some aspects seem to be relevant only for specific types of video tutorials:

- The tutorial should be visually appealing.
- The information in the tutorial should be up to date.
- The tutorial should not only describe what should be done, but also why it is done.
- The tutorial should also describe how to overcome potential problems if something goes wrong.

These points may be relevant for some tutorials, but more data are needed to draw a clear conclusion about their relevance.

We plan to use these results to further enhance existing approaches for the development of questionnaires for tutorial quality [8] with new scales.

This study provided first evidence that the perceived importance of quality criteria differs in principle between tutorial types. A natural next step is therefore to focus on specific subdomains of video tutorials and to investigate the observed effects there in more detail. Furthermore, the proposed classification and the quality criteria should be grounded more firmly in instructional theories, such as the Cognitive Theory of Multimedia Learning and the Cognitive Load Theory, and related to existing classifications of instructional videos.

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APPENDIX A

Email instructions per round (translated from German):

- Select one video from the three offered YouTube videos and watch it completely. All videos are only a few minutes long.
- Immediately after watching the video, complete the questionnaire. In the questionnaire, there is a field 'Please enter your code for assignment tracking' where you enter any code of your choice. Please use the identical code in all four questionnaires.
- Repeat these steps for all four tasks.

Important: You should not evaluate the quality of the video itself but rather assess which quality criteria would be important for evaluating tutorials of this type!

Links to the Tutorials (all retrieved April, 2026):

Group 1 (Assignment a):

- Cocktail-Mix-Tutorial (Mojito): https://www.youtube.com/watch?v=NkX_9dQn7DY
- Videos schneiden in Canva: https://www.youtube.com/watch?v=vbLR82alUoE
- Popcorn selber machen im Topf: https://www.youtube.com/watch?v=_tgomKwCPeE

Group 2 (Assignment b):

- Fotos professionell und schnell bearbeiten: https://www.youtube.com/watch?v=7X59I_pbqYg
- Die 4 Besten Teerezepte: https://www.youtube.com/watch?v=nUj_diOUKJA
- Geschenke einpacken für Anfänger: https://www.youtube.com/watch?v=QBx_2SqllSY

Group 3 (Assignment c):

- Einfaches Formular für Excel Tabelle erstellen: https://www.youtube.com/watch?v=TBFONS5TNfs
- Arbeiten mit dem Folienmaster – PowerPoint-Tutorial: https://www.youtube.com/watch?v=7K88oadVvOw
- Palstek | Seemannsknoten - Segeln lernen: https://www.youtube.com/watch?v=NpFb7nqA0Eg

Group 4 (Assignment d):

- Lebensmittel richtig schneiden wie ein Profi: https://www.youtube.com/watch?v=WP6BY5BxzEI
- Perspektive Zeichnen lernen: https://www.youtube.com/watch?v=Ob6HOCfnRQc
- Eigene Schriftart erstellen: https://www.youtube.com/watch?v=geG6-0QhBOs

APPENDIX B

Original German items on tutorial quality.

- I1: Wie wichtig ist es, dass ein Tutorial dieses Typs in klare, logische Schritte gegliedert ist.
- I2: Wie wichtig ist es, dass ein Tutorial dieses Typs zu Beginn klar die Voraussetzungen und die zu erwarteten Ergebnisse beschreibt.
- I3: Wie wichtig ist es, dass ein Tutorial dieses Typs visuell ansprechend gestaltet ist.
- I4: Wie wichtig ist es, dass ein Tutorial dieses Typs zur weiteren Beschäftigung mit dem Thema motiviert.
- I5: Wie wichtig ist es, dass ein Tutorial dieses Typs praktische Beispiele und angemessene Visualisierungen verwendet, um den Inhalt klar zu erklären.
- I6: Wie wichtig ist es, dass ein Tutorial dieses Typs leicht verständliche Erklärungen enthält.
- I7: Wie wichtig ist es, dass ein Tutorial dieses Typs ein für mich angemessenes Tempo hat.

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- I8: Wie wichtig ist es, dass ein Tutorial dieses Typs nur relevante Inhalte enthält und eine zufriedenstellende Dauer hat.
- I9: Wie wichtig ist es, dass die in einem Tutorial dieses Typs vermittelten Informationen auf dem neuesten Stand sind.
- I10: Wie wichtig ist es, dass ein Tutorial dieses Typs nicht nur erklärt, wie etwas gemacht wird, sondern auch warum es auf diese Weise gemacht wird.
- I11: Wie wichtig ist es, dass ein Tutorial dieses Typs auch mögliche Probleme beschreibt und Tipps zur Fehlerbehebung bietet, falls etwas schief geht.
- I12: Wie wichtig ist es, dass ein Tutorial dieses Typs motiviert zu experimentieren und eigene kreative Ideen zu entwickeln.
- I13: Wie wichtig ist es, dass der Lernende in einem Tutorial dieses Typs uninteressante Teile überspringen oder interessante Teile direkt ansteuern kann.
- I14: Wie wichtig ist es, dass die präsentierte Menge an Informationen in einem Tutorial dieses Typs zu jedem Zeitpunkt gut zu bewältigen ist.
- I15: Wie wichtig ist es, dass ein in einem Tutorial dieses Typs alle beschriebenen Schritte genau so funktionieren, wie sie im Tutorial beschrieben werden.
- I16: Wie wichtig ist es, dass ein Tutorial dieses Typs konkrete Übungsaufgaben anbietet, um das Gezeigte anzuwenden.
- I17: Wie wichtig ist es, dass ein Tutorial dieses Typs unmittelbares, hilfreiches Feedback zu Lösungen oder Schritten des Lernenden anbietet.

APPENDIX C

TABLE II. MEAN IMPORTANCE RATING PER QUALITY CRITERION AND TUTORIAL GROUP.

Item	Group 1	Group 2	Group 3	Group 4
I1	2.51	2.12	2.44	1.44
I2	1.60	1.19	1.66	1.41
I3	0.42	1.45	-0.37	1.00
I4	-0.53	-0.34	-0.49	0.66
I5	2.19	2.19	2.05	2.24
I6	2.23	2.19	2.58	1.98
I7	1.37	1.79	1.93	1.46
I8	1.74	1.02	2.12	1.49
I9	1.09	-1.05	0.78	-0.66
I10	-0.05	-0.12	1.29	0.61
I11	0.44	-0.50	1.02	0.15
I12	-1.19	-0.52	-1.29	0.24
I13	0.63	-0.07	0.39	0.34
I14	1.07	1.12	1.61	0.76
I15	2.19	1.79	2.73	1.18
I16	-1.63	-1.62	-0.76	-0.20
I17	-0.88	-0.93	-0.49	-0.73

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